

Crypto Homework 1

Submit the homework assignment on the course website. The solutions from pairs are accepted.

Submit a single file in zip format (only!) that contains:

- Source files and
- A WORD document led by students' names and ID numbers (should work in pairs only), followed, solutions and examples of input/output for every problem.

The symmetric block cipher **NameCipher** is given as follows:

The parameters:

Plane space: $X = \{0 \dots 25\}$

Cipher space: $Y = \{0 \dots 25\}$

Each block contains $m=2$ letters. $N = 26$;

Key space values: The key **K** is a $m \times m$ matrix whose elements are integers in Z_{26} .

The Encryption function defined by double encryption $Z = E_{K2}(E_{K1}(X))$ is given in the following way:

$Z = E_{K2}(E_{K1}(X)) \bmod N$, where $Y = E_{K1}(X) = (X * K_1 + (a,b)) \bmod N$, and

$Z = E_{K2}(Y * K_2 + (a,b)) \bmod N$, where a and b are integers in Z_{26} .

Comment: a plaintext (your names) is a string of the length between 4 and 10 characters; a key is a string given by a matrix 2×2 . For example: key = "road" = "r, o, a, d" = (17, 14, 0, 3);

$$K = \begin{pmatrix} r & o \\ a & d \end{pmatrix} = \begin{pmatrix} 17 & 14 \\ 0 & 3 \end{pmatrix}$$

Select your encryption keys K_1 and K_2 (for instance, $K_1 = \text{"road"}$ and $K_2 = \text{"door"}$).

Then, define the parameters a and b using the **first letters of your names** (for example, if the work is submitted by students Daniel and Razi, the parameters would be $a = 'D' = 3$ and $b = 'R' = 17$).

Exercise 1

1. Write a procedure in **Python** to encrypt a given plaintext by means of the cipher **NameCipher** with the **input:** plaintext and encryption key, **output:** the ciphertext.

The function prototype: **NameCipher_encryption (plaintext, key).**

Verify the selected encryption keys. Confirm the validity of the encryption keys and display all verification steps.

Encrypt your names. Provide 2 examples (encryption of your names) of the input/output data: plaintext, key and ciphertext.

Display the results in the Word file.

2. Write a procedure in **Python** to decrypt the received ciphertext.

The function prototype: **NameCipher_decryption (ciphertext, key).**

Decrypt the received ciphertext

Provide 2 examples (decryption of your names).
Display the results in the Word file.

Exercise 2

Write a procedure performing **the Iterative Attack** against the algorithm **NameCipher**.
Use the function **NameCipher_encryption** from Ex.1.1 to decrypt the received ciphertexts from Ex.1.1.
How many iterations does it take?
Give 2 examples (for your encrypted names) of input/output – ciphertext, encryption key, plaintext.

Exercise 3

What do you think?
For a pair of keys K_1 and K_2 , does there always exist a key K_3 that satisfies the condition
 $E_{K_2}(E_{K_1}(X)) = E_{K_3}(X)$ for each message X ? Justify your answer.
If yes, show which key K_3 replaces the pair of keys (K_1, K_2) from Exercise 1.

Good luck with your work!